

## APUTITEEQ, Greenland

WMO: 043510

Lat: 67.783N

Lon: 32.285W

Elev: 75

StdP: 14.66

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 3             | -5.7       | -1.4 | -14.8                         | 2.5 | -2.7 | -11.1 | 3.0 | 1.3  | 47.2                  | 26.5 | 40.1 | 20.0 | 5.6                   | 50   | 0.933 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.2                    | 47.2            | 41.2 | 45.2 | 40.1 | 43.5 | 39.2 | 42.4                | 45.2 | 41.1 | 43.8 | 40.0 | 42.6 | 7.4                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 40.5                            | 37.2 | 42.4 | 39.2 | 35.3 | 41.2 | 37.9 | 33.6 | 40.2 | 16.4          | 45.5 | 15.7 | 43.9 | 15.2 | 42.7 | 51.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 37.1          | 28.9 | 23.3 | DB                         | -9.4  | 52.5               | 6.4 | 2.4                                                | -14.0 | 54.2       | -17.8 | 55.6       | -21.4 | 57.0       | -26.0 | 58.7 | (4) |
| (5)               |      |      | WB                         | -10.5 | 45.6               | 6.2 | 2.5                                                | -15.0 | 47.4       | -18.6 | 48.9       | -22.1 | 50.3       | -26.6 | 52.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov  | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)  | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.0   | 18.7  | 16.0  | 15.1  | 23.0  | 30.1  | 33.4  | 37.5  | 38.1  | 36.0  | 30.1 | 25.4 | 19.8 |
|                                                                                  | DBStd       | 10.20  | 9.11  | 9.53  | 8.50  | 7.05  | 4.27  | 3.88  | 4.13  | 3.28  | 3.56  | 4.53 | 5.42 | 7.57 |
|                                                                                  | HDD50       | 8395   | 970   | 951   | 1083  | 810   | 616   | 497   | 388   | 369   | 420   | 616  | 738  | 937  |
|                                                                                  | HDD65       | 13870  | 1435  | 1371  | 1548  | 1260  | 1081  | 947   | 853   | 834   | 870   | 1081 | 1188 | 1402 |
|                                                                                  | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0    |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0    |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0    |
| Wind                                                                             | WSAvg       | 7.9    | 10.1  | 9.5   | 8.9   | 8.1   | 6.7   | 5.3   | 5.4   | 5.7   | 7.2   | 9.1  | 9.3  | 9.2  |
| Precipitation                                                                    | PrecAvg     | 36.3   | 3.9   | 3.8   | 3.5   | 2.7   | 2.0   | 2.1   | 1.4   | 2.9   | 3.6   | 3.2  | 3.7  | 3.5  |
|                                                                                  | PrecMax     | 51.4   | 8.7   | 11.3  | 7.2   | 5.8   | 4.9   | 4.9   | 3.3   | 4.3   | 9.4   | 5.2  | 9.9  | 10.0 |
|                                                                                  | PrecMin     | 23.5   | 1.3   | 0.9   | 0.7   | 0.7   | 0.6   | 0.9   | 0.3   | 1.1   | 1.0   | 1.5  | 0.6  | 0.4  |
|                                                                                  | PrecStd     | 7.8    | 2.0   | 2.4   | 1.8   | 1.3   | 0.9   | 0.8   | 0.9   | 0.9   | 1.8   | 1.0  | 2.1  | 2.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 39.1  | 35.8  | 34.2  | 42.0  | 43.1  | 45.7  | 52.1  | 48.6  | 47.5  | 43.5 | 39.6 | 37.7 |
|                                                                                  |             | MCWB   | 31.8  | 31.6  | 29.1  | 37.3  | 36.3  | 39.9  | 44.7  | 40.8  | 40.5  | 37.0 | 33.6 | 31.3 |
|                                                                                  | 2%          | DB     | 34.2  | 33.2  | 31.6  | 37.1  | 40.2  | 43.1  | 48.1  | 46.2  | 44.0  | 40.0 | 35.6 | 34.5 |
|                                                                                  |             | MCWB   | 30.6  | 30.8  | 28.8  | 34.2  | 35.3  | 38.5  | 42.6  | 40.7  | 38.4  | 36.5 | 32.6 | 30.2 |
|                                                                                  | 5%          | DB     | 32.2  | 31.6  | 29.7  | 34.6  | 37.8  | 41.2  | 45.7  | 44.5  | 42.0  | 38.2 | 33.6 | 31.8 |
|                                                                                  |             | MCWB   | 29.9  | 30.1  | 27.3  | 32.5  | 34.1  | 37.5  | 41.0  | 39.9  | 37.9  | 35.4 | 31.4 | 29.5 |
|                                                                                  | 10%         | DB     | 30.3  | 29.1  | 27.1  | 33.0  | 35.9  | 39.4  | 43.6  | 42.9  | 40.5  | 36.3 | 32.1 | 29.8 |
|                                                                                  |             | MCWB   | 28.2  | 27.2  | 25.1  | 31.4  | 33.1  | 36.2  | 39.8  | 39.2  | 37.3  | 33.7 | 30.3 | 27.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 33.5  | 33.2  | 31.5  | 37.7  | 37.8  | 41.1  | 47.0  | 43.7  | 42.3  | 39.2 | 35.0 | 33.7 |
|                                                                                  |             | MCDB   | 35.0  | 34.3  | 32.6  | 40.3  | 40.7  | 44.5  | 50.0  | 46.1  | 43.8  | 41.0 | 37.7 | 35.9 |
|                                                                                  | 2%          | WB     | 32.1  | 31.9  | 29.6  | 34.7  | 36.0  | 39.2  | 43.4  | 41.8  | 40.5  | 37.6 | 33.3 | 31.6 |
|                                                                                  |             | MCDB   | 33.8  | 32.7  | 31.0  | 36.5  | 38.8  | 42.0  | 46.9  | 44.2  | 42.1  | 39.2 | 34.7 | 32.8 |
|                                                                                  | 5%          | WB     | 30.3  | 29.9  | 27.5  | 33.2  | 34.7  | 37.8  | 41.5  | 40.7  | 39.4  | 36.0 | 32.1 | 29.7 |
|                                                                                  |             | MCDB   | 31.7  | 31.4  | 29.4  | 34.4  | 37.0  | 40.5  | 44.8  | 43.2  | 40.9  | 37.5 | 33.0 | 31.4 |
|                                                                                  | 10%         | WB     | 28.3  | 27.4  | 25.1  | 31.8  | 33.6  | 36.6  | 40.1  | 39.8  | 38.2  | 34.1 | 30.8 | 27.8 |
|                                                                                  |             | MCDB   | 30.1  | 28.9  | 26.9  | 32.9  | 35.4  | 39.0  | 43.2  | 42.2  | 39.8  | 35.8 | 32.0 | 29.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.2   | 10.1  | 11.0  | 9.9   | 8.1   | 7.9   | 8.3   | 7.2   | 6.1   | 5.1  | 5.5  | 7.7  |
|                                                                                  |             | MCDBR  | 9.4   | 9.3   | 11.2  | 10.5  | 10.7  | 10.2  | 11.4  | 9.4   | 8.4   | 7.1  | 7.7  | 8.8  |
|                                                                                  |             | MCWBR  | 7.4   | 7.3   | 9.4   | 7.5   | 6.9   | 6.6   | 6.9   | 5.6   | 5.1   | 5.0  | 5.5  | 7.3  |
|                                                                                  | 5% WB       | MCDBR  | 9.4   | 9.0   | 10.7  | 9.0   | 9.1   | 9.2   | 10.2  | 8.3   | 7.1   | 6.2  | 6.8  | 8.3  |
|                                                                                  |             | MCWBR  | 8.7   | 7.8   | 9.3   | 6.6   | 6.4   | 6.3   | 7.0   | 5.6   | 5.1   | 4.7  | 5.3  | 7.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.202  | 0.229 | 0.267 | 0.313 | 0.320 | 0.320 | 0.322 | 0.328 | 0.304 | 0.256 | N/A  | N/A  |      |
|                                                                                  | taud        | 1.837  | 2.049 | 2.143 | 2.039 | 2.073 | 2.202 | 2.352 | 2.376 | 2.341 | 2.120 | N/A  | N/A  |      |
|                                                                                  | Ebn at Noon | 29     | 185   | 246   | 261   | 274   | 278   | 275   | 259   | 233   | 166   | N/A  | 0    |      |
|                                                                                  | Edh at Noon | 3      | 16    | 26    | 39    | 42    | 38    | 32    | 28    | 22    | 14    | N/A  | 0    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 12     | 170   | 690   | 1445  | 2169  | 2376  | 2023  | 1394  | 717   | 258   | 33   | 0    |      |
|                                                                                  | RadStd      | 1      | 23    | 43    | 85    | 103   | 143   | 114   | 71    | 84    | 27    | 3    | 0    |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page